

Education

August 2022 **McGill University**
– present Bachelor of Science - Honours Mathematics.
GPA: 3.81/4.0

Montréal, Canada

Research Experience

May 2025 **Research Project on Special Values of L-functions**

– present Supervisor: Professor Henri Darmon (McGill University)

- Started writing an expository paper on the detailed history of L-functions and the mathematicians that contributed to that subject: [pdf (unfinished)]
- Funded by the NSERC (Natural Sciences and Engineering Research Council of Canada).

May 2024 **Research Project on the Ergodic Theorems**

– August 2024 Supervisor: Professor Anush Tserunyan (McGill University)

- Wrote an expository paper on the ergodic theorems with a focus on Pr. Tserunyan's new proof of the Birkhoff Ergodic Theorem: [pdf]

January 2024 **Directed Reading Program on p -adic Numbers**

– April 2024 Supervisor: Doctoral student Hazem Hassan (McGill University)

- Studied p -adic numbers and p -adic Zeta-Functions by reading the book *p -adic Numbers, p -adic Analysis, and Zeta-Functions* by Neal Koblitz.

Professional Experience

September 2025 **Teaching Assistant for the course History & Philosophy of Mathematics**
– December 2025 *McGill University, Montréal, QC*

Instructor: Professor Henri Darmon, Course Code: MATH 338, 135 students

January 2024 **Grader for the course Discrete Structures**

– April 2024 *McGill University, Montréal, QC*

Instructor: Mr. Jeremy Macdonald, Course Code: MATH 240, 365 students

May 2023 **Tutor in Mathematics**

September 2024 *McGill University, Montréal, QC*

Projects

January 2025 **Expository Paper on the Divergence of the Harmonic Series**

- Wrote a self-contained paper presenting the importance of rigor in mathematics through the detailed study of the divergence of the Harmonic Series: [pdf]

December 2024 **Published Historical Article on Fourier Analysis**

- Co-authored (along with Nisrine Sqalli) a historical paper presenting our findings: [pdf]
- Published a condensed version in the Delta Epsilon journal: [pdf]

December 2024 **Personal Website**

- Coded a personal website by learning and using HTML and CSS: [samylahlou.com]

Awards

NSERC Undergraduate Student Research Award – \$8950

Languages

English (Fluent)

French (Native)

Arabic (Native)